

Benchmarking ERP Analysis: Manual Features, Deep Learning, and Foundation Models

Yihe Wang, Zhiqiao Kang,* Bohan Chen,* Yu Zhang, Xiang Zhang✉

Abstract—Event-related potential (ERP), a specialized paradigm of electroencephalographic (EEG), reflects neurological responses to external stimuli or events, generally associated with the brain’s processing of specific cognitive tasks. ERP plays a critical role in cognitive analysis, the detection of neurological diseases, and the assessment of psychological states. Recent years have seen substantial advances in deep learning-based methods for spontaneous EEG and other non-time-locked task-related EEG signals. However, their effectiveness on ERP data remains underexplored, and many existing ERP studies still rely heavily on manually extracted features. In this paper, we conduct a comprehensive benchmark study that systematically compares traditional manual features (followed by a linear classifier), deep learning models, and pre-trained EEG foundation models for ERP analysis. We establish a unified data preprocessing and training pipeline and evaluate these approaches on two representative tasks across 12 datasets: ERP stimulus classification and ERP-based brain disease detection. Furthermore, we investigate various patch-embedding strategies within advanced Transformer architectures to identify embedding designs that better suit ERP data. Our study provides a landmark framework to guide method selection and tailored model design for future ERP analysis. The code is available at <https://github.com/DL4mHealth/ERP-Benchmark>

Index Terms—ERP, event-related potential, EEG, brain-computer interfaces, deep learning, foundation model

I. INTRODUCTION

Event-related potential (ERP) is an EEG-derived measure of time-locked neural activity, serving as a foundational tool with broad applications in neuroscience, dementia, and cognitive science [1]–[3]. As a specialized category within the broader EEG signal domain, ERP exhibits distinct characteristics compared to spontaneous EEG activity [4]. While spontaneous EEG reflects ongoing intrinsic brain states but with limited functional specificity, ERP captures time-locked neurological responses elicited by specific sensory, audio, or cognitive events [5]–[7].

Yihe Wang and Xiang Zhang are with the Department of Computer Science, University of North Carolina-Charlotte, Charlotte, North Carolina 28223, United States.

Zhiqiao Kang is with the College of Future Technology, South China University of Technology, Guangzhou, Guangdong 511442, China.

Bohan Chen is with the Department of Statistics, University of California-Davis, Davis, California 95618, United States.

Yu Zhang is with the Department of Public Mental Health & Population Sciences, Stanford University, Palo Alto, California 94305, United States.

Xiang Zhang is the corresponding author (E-mail: xiang.zhang@charlotte.edu).

*Equal Contribution. This work was conducted during their research internship at UNC Charlotte.

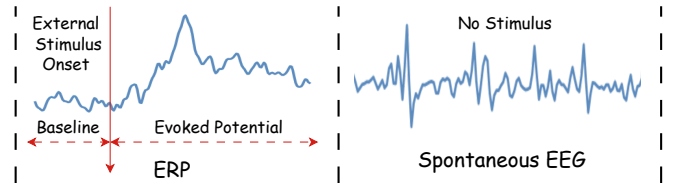


Fig. 1: Comparison Between ERP and Spontaneous EEG.

By aligning neural signals to controlled stimuli, ERP provides a direct and interpretable window into transient cognitive processes. The primary advantage of ERP lies in its temporal synchrony between external events and internal brain electrical responses, which typically yields a higher signal-to-noise ratio than spontaneous EEG activity [8]. In practice, ERP trials are extracted from continuous EEG recordings by segmenting event-centered epochs [3]. A short pre-stimulus interval (typically 0.2-0.5 seconds) is used for baseline correction to suppress background noise, followed by a longer post-stimulus interval (typically 0.8-1.5 seconds) that contains the stimulus-evoked neural potential [9]. Figure 1 illustrates a comparison between ERP and spontaneous EEG.

In recent years, deep learning has achieved substantial progress in EEG decoding across a wide range of applications, including sleep stage classification [10], seizure detection [11], and dementia detection [12]. Advanced deep learning techniques, such as foundation models that have demonstrated remarkable success in natural language processing, computer vision, and multimodal learning (e.g., ChatGPT [13], Gemini [14], and DINOv2 [15]), are also beginning to be explored for learning generalizable EEG representations. Recent EEG foundation models, including LaBraM [16], EEGPT [17], and CBraMod [18], have reported strong performance on various EEG downstream tasks. However, most existing deep learning model designs primarily target spontaneous EEG signals, such as resting-state and sleep EEG [10], or non-time-locked task-induced EEG, such as motor imagery [19]. In contrast, time-locked evoked EEG signals, exemplified by ERP, remain significantly underexplored. As a result, ERP analysis continues to rely mainly on manual feature extraction and conventional statistical approaches, or directly employs models designed for spontaneous EEG to ERP analysis, resulting in suboptimal performance. Given the distinct signal characteristics of ERP and its fundamental differences from spontaneous EEG, it remains unclear to what extent recent advances in deep learning are directly applicable to ERP decoding. This gap necessitates systematic benchmarking of methods for ERP analysis.

To this end, we conduct a comprehensive and fair benchmark

evaluation of ERP classification methods. We implement a wide range of approaches, including 31 widely used EEG features and 91 ERP features, 10 state-of-the-art deep learning methods for EEG, and 3 advanced EEG foundation models, within a unified data preprocessing and evaluation framework. The benchmark covers two representative and practical ERP classification paradigms across **12 datasets**. The first focuses on **ERP stimulus classification**, such as distinguishing standard and target stimuli in oddball paradigms, to evaluate models' ability to capture fundamental stimulus-locked cognitive responses. The second addresses **ERP-based brain disease detection**, such as identifying neurological or psychiatric conditions, thereby assessing model utility in more complex and clinically relevant scenarios. Furthermore, to guide the development of more effective models, we conduct a comparative analysis of three prevalent patch-embedding strategies for EEG data within a state-of-the-art Transformer framework. This systematic comparison seeks to determine the optimal embedding method for ERP classification, providing critical insights for designing dedicated Transformer backbones for ERP analysis.

In summary, our motivation is to answer the following question through this systematic benchmark study:

- **Q1:** *How do deep learning-based EEG methods compare with traditional manual feature extraction methods when applied to ERP data?*
- **Q2:** *Do EEG Foundation Models with pre-trained weights outperform supervised EEG deep learning models trained from scratch on ERP classification tasks?*
- **Q3:** *What is the most robust and generalizable method for ERP classification so far?*
- **Q4:** *Which patch embedding strategy is most suitable for designing ERP-specific Transformer models?*

To provide a concise takeaway message, we summarize the answers below for the reader's convenience:

- **A1:** *Most deep learning methods consistently outperform manual feature extraction approaches on ERP tasks.*
- **A2:** *Existing EEG foundation models do not demonstrate clear performance advantages over supervised deep learning models trained from scratch.*
- **A3:** *EEGConformer achieves best average ranking and demonstrates the most competitive overall performance.*
- **A4:** *Uni-variate patch embedding achieves better overall performance than multi-variate and whole-variate patching strategies for ERP-specific Transformer.*

The rest of this paper is organized as follows: Section III presents statistical information about the ERP datasets used in this study, along with the unified preprocessing pipeline. Section IV describes the methods compared in our benchmark, as well as the test Transformer implemented for embedding comparison. Section V provides implementation details and reports experimental results with corresponding analysis. Section VI discusses the limitations of this study and outlines directions for future work. Section VII concludes the paper.

II. RELATED WORK

Manual Extract Features: This method aims to identify potential biomarkers in ERP signals for classification or brain

interpretation. Different types of features are used, including statistical features like mean, skewness, kurtosis, and standard deviation [20]–[22], time-domain shape features such as line length, peak-to-peak amplitude, and positive/negative peak amplitudes [23]–[25], spectral distribution features like phase shift, phase coherence, bispectrum, and bicoherence [26]–[29], frequency-band power features like power spectrum density, relative band power, ratio of EEG rhythm, and energy [30]–[32], complexity features like Shannon entropy, Tsallis entropy, and permutation entropy [33]–[35], and Hjorth parameters such as activity, mobility, and complexity [36]–[38].

Deep Learning Methods: Deep learning methods with diverse architectures are widely used for EEG decoding. EEGNet [39] is a classic lightweight convolutional model, while EEG-Inception [40] extends temporal convolution with depthwise spatial blocks for ERP-based BCIs. AMWCNN [41] integrates wavelet-based multiscale analysis and attention mechanisms to identify discriminative frequency bands and temporal segments. Transformer-based approaches, such as EEG-Transformer [42] and EEGConformer [43], leverage self-attention to capture long-range temporal dependencies and global contextual information in EEG signals. In addition, Neuro-BERT [44] introduces masked autoencoding for robust EEG representation learning, LGGNet [45] explicitly models local–global brain functional graphs, MOCNN [46] proposes multiscale octave convolution for ERP classification, and EEGMamba [47] employs bidirectional Mamba modules for efficient long-sequence EEG modeling.

Foundation Models: Foundation models, which have achieved remarkable success in computer vision and natural language processing domains, are increasingly being explored for EEG representation learning. LaBraM [16] introduces a large-scale EEG foundation model that employs a neural tokenizer to reconstruct spectral representations during pre-training, enabling effective transfer to downstream EEG classification tasks. EEGPT [17] proposes a masked self-supervised learning framework based on high signal-to-noise EEG representations, enhanced by spatio-temporal representation alignment. NeuroLM [48] conceptualizes EEG signals as a foreign language and adopts language-model-inspired architectures to learn unified representations across cognitive and clinical tasks. LUNA [49] presents a topology-agnostic EEG foundation model using cross-attention mechanisms, supporting robust learning under heterogeneous channel montages. CSBrain [50] further explores cross-scale spatiotemporal tokenization with structured sparse attention to model dependencies between local and global brain regions.

III. DATASETS

A. Datasets

In light of the limited availability of public ERP datasets, we systematically searched all accessible resources known to the authors, encompassing platforms such as OpenNeuro, FigShare, and PhysioNet. We identified 12 datasets, each with a sufficient sample size (40+ subjects) to ensure robust statistical evaluation, including **ADHD-WMRI** [51], **AOPD** [52], **CESCA-AOOD**, **CESCA-VOOD**, **CESCA-FLANKER** [53],

TABLE I: **Dataset Statistics.** All the datasets follow the same preprocessing pipeline, including a band-pass filter, artifact removal, and resampling to 200Hz. The epoch and baseline window depend on the ERP tasks and datasets. Abbreviations: **AODD**: Auditory Oddball; **VODD**: Visual Oddball; **MSIT+**: Extended multi-source interference task; **SIM**: Simon Conflict; **RL**: Reinforcement Learning; **HC**: Healthy Control; **PD**: Parkinson’s Disease; **ADHD**: Attention Deficit Hyperactivity Disorder.

Datasets	ERP Task	#Subjects	Baseline(s)	Epoch(s)	#Trials	#Channels	Classes
CESCA-AODD	AODD	127	[-0.2, 0]	[-0.2, 0.8]	38,151	26	Standard vs Target
CESCA-VODD	VODD	127	[-0.2, 0]	[-0.2, 0.8]	20,419	26	Standard vs Target
CESCA-FLANKER	Flanker	73	[-0.2, 0]	[-0.2, 0.8]	29,774	26	Congruent vs Incongruent
mTBI-ODD	AODD	96	[-0.2, 0]	[-0.2, 0.8]	24,885	61	Standard vs Target vs Novel
NSERP-MSIT	MSIT+	42	[-0.5, 0]	[-0.5, 1.0]	16,729	123	Non-Conflict vs Simon Effect vs Flanker Effect vs Double-Conflict
NSERP-ODD	VODD	42	[-0.5, 0]	[-0.5, 1.0]	27,865	123	Standard vs Target vs Novel
PD-SIM	SIM	147	[-0.3, -0.2]	[-0.5, 1.0]	55,921	60	HC vs PD
PD-ODD	VODD	145	[-0.3, -0.2]	[-0.5, 1.0]	34,464	60	HC vs PD
ADHD-WMRI	N-Back, GoNogo	59	[-0.2, 0]	[-0.2, 0.65]	21,832	21	HC vs ADHD
SCPD	SIM	56	[-0.3, -0.2]	[-0.5, 1.0]	10,224	59	HC vs PD
RLPD	RL	56	[-0.2, 0]	[-2.0, 1.0]	14,325	56	HC vs PD
AOPD	AODD	50	[-0.2, 0]	[-0.2, 0.8]	9,830	59	HC vs PD

mTBI-ODD [54], NSERP-MSIT, NSERP-ODD [55], PD-ODD [56], PD-SIM [56], RLPD [57], and SCPD [58]. These datasets span a broad range of cognitive paradigms and clinical conditions, including attention deficit hyperactivity disorder (ADHD), Parkinson’s disease (PD), traumatic brain injury (TBI), and aging-related cognitive decline. To ensure fair and consistent benchmarking, all datasets are preprocessed using a unified ERP preprocessing pipeline (Sec. III-B).

Among them, 6 datasets, including CESCA-AODD, CESCA-VODD, CESCA-FLANKER, mTBI-ODD, NSERP-MSIT, and NSERP-ODD, are used for ERP task classification, which aims to distinguish cognitive responses elicited by different experimental conditions, such as target versus non-target and congruent versus incongruent stimuli. The remaining 6 datasets, including PD-SIM, PD-ODD, RLPD, SCPD, AOPD, and ADHD-WMRI, are employed for brain disease detection tasks.

B. Data Preprocessing Pipeline

To utilize ERP trials for training, we apply the following unified preprocessing pipeline. **1) Removal of non-EEG channels:** All non-EEG channels are removed, such as EOG or coordinate information. **2) Notch and band-pass filtering:** A notch filter at 50 Hz or 60 Hz is applied to suppress line noise, followed by a band-pass filter between 0.5 Hz and 45 Hz. **3) Bad channel interpolation:** Bad channels are interpolated when marked. **4) Average re-referencing:** Average re-referencing is applied to reduce global noise and potential baseline shifts. **5) Artifact removal:** Independent component analysis (ICA), combined with ICLabel [59], is used to automatically identify and remove components associated with eye blinks, muscle activity, and cardiac artifacts. **6) Resampling:** All recordings are resampled to a uniform sampling rate of 200 Hz. **7) Baseline correction:** Baseline correction is applied to enhance stimulus-related ERP components relative to background noise. **8) Trial epoching:** EEG recordings are then segmented into ERP trials based on stimulus events, and the non-ERP recordings are discarded (e.g., resting state). The baseline correction and epoch window length vary across ERP tasks and datasets. **9) Z-Score Normalization:** Each channel of every segmented ERP trial is normalized independently to zero mean and unit variance. The statistics of

the processed datasets are summarized in Table I. More details of the preprocessing are provided in our GitHub repository.

IV. METHOD

A. Manual Extract Features

Widely used features in spontaneous EEG and ERP data analysis. We manually extract these features and apply a linear projection layer to them for ERP task classification.

1) EEG Feature Extraction: We compute **31 widely used features**, including 10 time-domain statistical features, 11 frequency band features, 7 spectral distribution features, and 3 data complexity features.

Time-Domain Statistical Features: These include the mean, median, minimum, maximum, skewness, kurtosis, root mean square (RMS), interquartile range, and standard deviation [20]–[22], [32]. Statistical features are designed to capture fundamental time-domain statistical characteristics of EEG signals.

Frequency-Band Power Features: This category consists of absolute power in the δ , θ , α , and β bands, total power, the θ/α ratio, the α/β ratio, and the relative power of the four frequency bands [30]–[32]. Absolute band power is computed by numerically integrating the power spectral density (PSD) within each frequency band, providing a direct measure of oscillatory energy associated with specific neuro processes.

Spectral Distribution Features: The spectral features include spectral centroid, spectral roll-off frequency, peak frequency, peak power, mean frequency, median frequency, and spectral flatness [60]–[62]. These features characterize the overall distribution and structure of spectral energy. All spectral features are estimated using Welch’s method for PSD estimation, followed by numerical integration and computations.

Complexity features: This category includes normalized & non-normalized Shannon spectral entropy and Tsallis entropy [33]–[35]. These features are derived from the normalized PSD, which is interpreted as a probability distribution. Shannon spectral entropy quantifies spectral disorder and complexity, with higher values indicating more uniformly distributed spectral energy, while Tsallis entropy generalizes Shannon entropy by incorporating non-extensive properties, providing greater sensitivity to nonlinear and non-stationary signal characteristics.

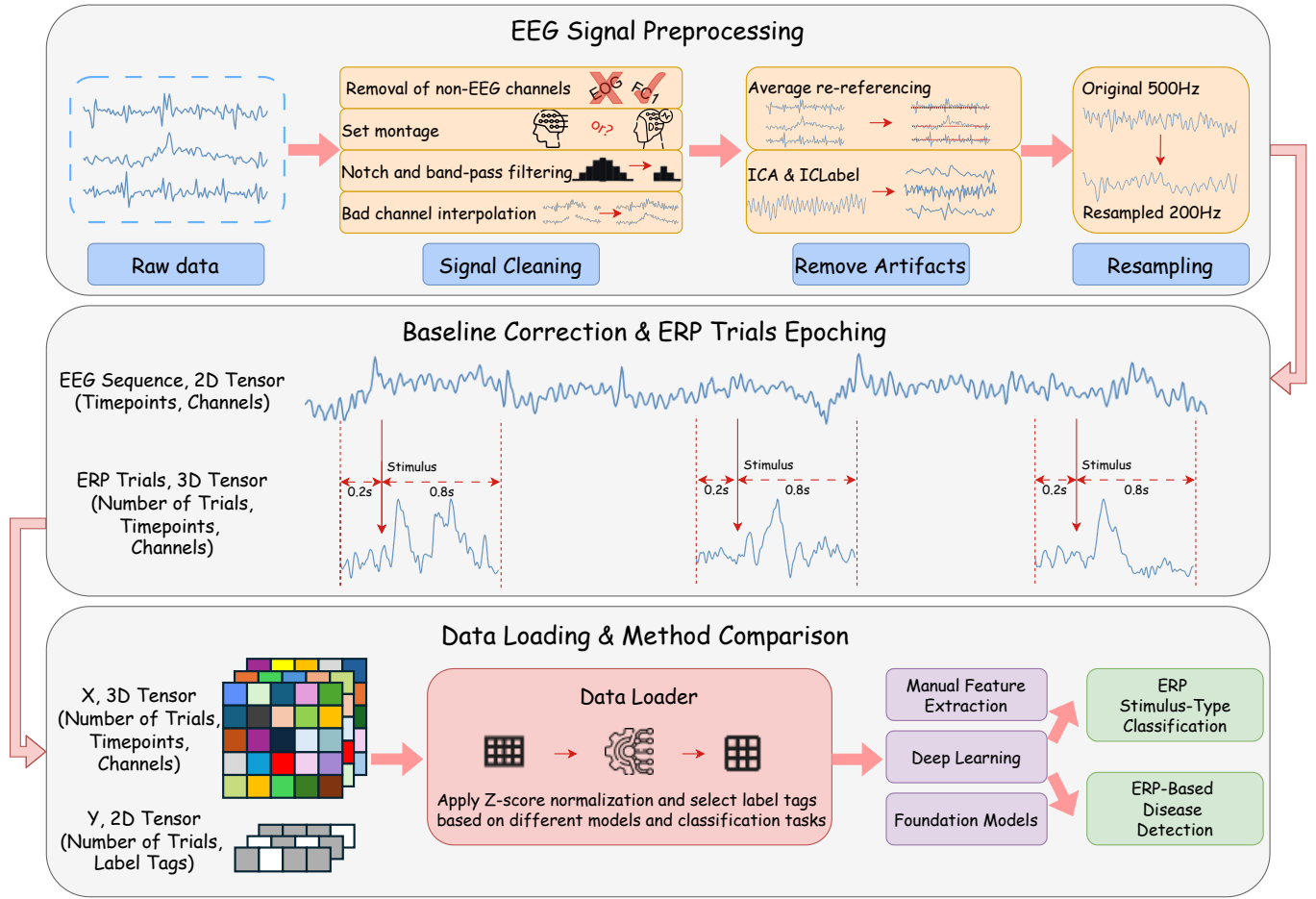


Fig. 2: **Pipeline of ERP Analysis.** Input raw EEG data are preprocessed with a unified pipeline to get ERP trials, including removal of non-EEG channels, notch and band-pass filtering, bad channel interpolation, average re-referencing, artifact removal, resampling, baseline correction, trial epoching, and Z-score normalization. The processed ERP trials are loaded and passed to various models for training and classification, including manual feature extraction, supervised deep learning trained from scratch, and foundation models with pre-trained weights.

2) ERP Feature Extraction: We compute a total of **91 features**, including 75 temporal pyramid pooling features, 4 peak-related Features, 7 frequency-band & complexity features, and 3 Hjorth parameters.

Temporal Pyramid Pooling: five statistical measures, including mean, standard deviation, RMS, line length, and peak-to-peak amplitude [63]–[66], are computed across multiple temporal scales, yielding $5 \times \sum$ levels feature dimensions. We set the number of levels equal to 15. Temporal Pyramid Pooling is employed to capture ERP dynamics at multiple temporal resolutions. This approach jointly captures global waveform structure and localized transients, improving robustness to temporal variability and latency jitter.

Peak-Related Features: amplitudes of positive and negative peaks and their normalized latencies [23]–[25]. Peak-related features are extracted to characterize salient ERP components. The maximum positive and minimum negative peak amplitudes reflect the strength of event-related neural responses. The total analysis window length-normalized latencies to account for variability in ERP duration and ensure comparability across trials and subjects.

Frequency & Complexity Features (9): relative power in the δ , θ , α , and β bands, total power, spectral centroid, spectral flatness, median frequency, and normalized spectral entropy [26]–[29]. Although ERPs are time-locked, trial-averaged signals, frequency-domain analysis provides complementary information about oscillatory patterns elicited by stimuli. Relative band power and total power summarize energy distribution across canonical frequency bands, while spectral centroid, spectral flatness, and median frequency describe the overall frequency structure of the ERP signal. Normalized spectral entropy is computed to quantify the complexity and dispersion of ERP spectral content, with lower entropy indicating more concentrated frequency responses associated with event-related processing.

Hjorth parameters (3): Hjorth parameters includes activity, mobility, and complexity [36]–[38]. These features serve as compact descriptors of ERP signal characteristics. Activity reflects the overall signal amplitude by measuring signal variance. Mobility characterizes the mean frequency content by relating the variance of the first derivative to the variance of the signal itself. Complexity quantifies waveform shape variation

relative to a sinusoidal signal, thereby providing insight into the structural intricacy of ERP dynamics.

B. Deep Learning Models

Deep learning methods for general time-series, medical time-series, and EEG data classification. They are trained from scratch in a fully supervised learning manner. We select classic or state-of-the-art methods with publicly available code.

1) *TCN*: TCN [67] is a convolutional architecture designed explicitly for time-series modeling. They extend residual networks by incorporating causal and dilated convolutions, enabling effective learning of long-range temporal dependencies. By flexibly controlling receptive field size, TCNs balance model capacity and computational efficiency, thereby making them adaptable to time-series data across diverse domains. TCNs remain a strong and widely adopted baseline for time-series classification tasks, including EEG analysis.

2) *ModernTCN*: ModernTCN [68] is a convolutional architecture designed for general-purpose time-series analysis and demonstrates strong performance, particularly in classification. It combines depthwise convolutions with multiple pointwise convolutions and introduces channel-independent embedding mechanisms. This design effectively addresses key limitations of convolutional models in multivariate time-series tasks, where performance is often constrained by small receptive fields and difficulties in modeling inter-variable dependencies.

3) *TimesNet*: TimesNet [69] is a state-of-the-art model for time-series analysis, especially on classification tasks. It transforms one-dimensional time-series data into two-dimensional tensors across multiple periodicities (where rows and columns represent intra- and inter-period variations, respectively), thereby leveraging the strengths of 2D convolutional networks for time-series analysis. The model's core component, TimesBlock, employs a parameter-efficient Inception module to extract complex temporal variations from 2D tensors.

4) *PatchTST*: PatchTST [70] is a transformer-based model for time series prediction and representation learning, with its core innovation lying in a dual design of patching and channel independence. The model decomposes multi-channel data into multiple one-channel patches, treating them as semantically informative sub-sequence fragments for input. This approach preserves local features while significantly reducing computational complexity.

5) *iTransformer*: The iTransformer [71] proposes a novel data-embedding method based on the transformer architecture for multivariate time-series analysis. Its distinctive feature lies in reversing the conventional tokenization strategy. Instead of merging multi-channel samples at a single time point into a single temporal token, iTransformer treats each time series as an independent variable token. This adjustment enables the self-attention module to model correlations between variables directly, thereby enhancing interpretability.

6) *Medformer*: Medformer [72] is a multi-granularity patching Transformer model designed explicitly for medical time-series data (e.g., EEG, ECG). It captures correlations across signal channels via cross-channel patching and extracts features at multiple temporal scales via multi-granularity embedding.

Subsequently, it applies two-stage (intra-granularity & inter-granularity) multi-granularity self-attention to learn features. Medformer is a modified Transformer model adapted for medical electro-biological time-series data.

7) *MedGNN*: MedGNN [73] proposes a multi-scale spatio-temporal graph learning framework for medical time series such as electroencephalograms. This framework models cross-channel spatial relationships by constructing adaptive graph structures and integrating multi-scale temporal features via frequency-domain convolution. MedGNN further employs a differential-based attention mechanism and multi-scale graph transformers to capture local temporal dynamics and global graph dependencies in a synergistic manner.

8) *EEGNet*: As a classic deep learning application in the EEG domain, EEGNet [39] efficiently extracts temporal and spatial features via convolutional operations. By stacking 2D temporal, depthwise, and separable convolution modules, it captures frequency features and feature relationships across different electrode channels. Its structure is compact and requires low space and computational cost.

9) *EEGInception*: EEGInception [40] pioneers the integration of Inception modules [74] into deep learning models for ERP detection tasks. By employing multi-scale temporal convolutions to extract features across different temporal scales within EEG signals concurrently, and incorporating separable convolutions, batch normalization, and Dropout, it constructs a lightweight and efficient network architecture.

10) *EEGConformer*: EEGConformer [75] is a Transformer-based architecture that integrates convolutional networks for spatio-temporal feature embedding with self-attention mechanisms for EEG representation learning. This design combines the strengths of convolutional networks in capturing local spatio-temporal patterns and self-attention in modeling global dependencies, while maintaining parameter efficiency and enhancing the ability to represent complex EEG structures.

C. Foundation Models

Medical time-series and EEG foundation models. We use their pre-trained weights for fine-tuning on our ERP tasks. We select state-of-the-art methods with publicly available code.

1) *BIOT*: [76] BIOT introduces a large foundation model for the flexible processing of multi-channel biomedical signals of varying lengths and channels. It adopts language-model-style segmentation to convert single-channel signals into patch embeddings, augments them with channel tags, and applies self-attention to capture temporal and spatial dependencies. This design supports multi-stage training and remains effective under missing-channel or missing-segment scenarios. We load their pre-trained weights for fine-tuning.

2) *LaBraM*: LaBraM [16] is a foundation model for learning universal EEG representations through large-scale unsupervised pre-training. The framework's core lies in a vector-quantization neural spectral prediction method: First, a neural tokenizer is trained to encode continuous EEG segments into a compact neural lexicon by reconstructing the Fourier spectrum of the raw EEG signals. Subsequently, a neural Transformer is pre-trained using masked EEG modeling (randomly masking portions of

TABLE II: **ERP Stimulus Classification Results.** Classifying ERP stimulus types such as target, non-target, and distractor in Oddball, or congruent and incongruent in Flanker. **Top-1**, **Top-2**, and **Top-3** results are highlighted in red, blue, and green.

Datasets	CESCA-AODD (38,151 Trials) (127 Subjects, 2 Classes)			CESCA-VODD (20,419 Trials) (127 Subjects, 2 Classes)			CESCA-FLANKER (29,774 Trials) (73 Subjects, 2 Classes)		
Metrics	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC
EEG Features	<u>77.06±0.96</u>	46.44±1.03	50.71±0.38	77.26±1.02	50.31±0.84	56.73±1.16	55.38±1.12	55.13±1.12	57.74±1.84
ERP Features	75.43±3.46	47.74±1.66	52.35±1.67	<u>81.71±0.43</u>	63.68±1.10	76.67±1.83	<u>64.12±1.31</u>	63.98±1.23	69.66±1.57
TCN	75.07±0.40	50.74±0.99	56.39±0.95	82.03±0.69	67.00±1.12	<u>78.13±1.42</u>	63.16±1.11	62.89±1.30	68.62±1.43
ModernTCN	75.20±0.69	53.93±0.82	60.36±0.98	<u>81.91±0.51</u>	66.85±3.11	<u>78.18±1.01</u>	64.07±0.58	<u>64.01±0.59</u>	<u>69.90±1.39</u>
TimesNet	75.17±0.58	54.38±1.34	60.91±0.74	81.13±1.53	<u>67.65±1.55</u>	77.62±1.42	63.20±0.64	63.09±0.59	68.29±1.25
PatchTST	<u>76.41±1.41</u>	53.65±1.70	<u>61.98±3.11</u>	80.97±1.13	<u>68.19±1.72</u>	78.13±1.02	63.81±0.77	63.64±0.81	69.61±1.33
iTransformer	74.98±1.22	53.58±0.63	60.30±1.21	81.39±0.49	66.20±1.12	76.37±0.86	63.77±1.22	63.69±1.19	69.35±1.68
Medformer	74.41±0.90	57.18±0.42	62.87±1.06	80.25±1.02	66.58±0.88	75.85±1.23	63.56±0.98	63.42±1.06	69.26±1.41
MedGNN	74.68±0.38	<u>55.29±0.77</u>	60.92±0.68	81.55±0.86	67.52±1.03	77.95±1.16	64.43±1.21	64.33±1.22	70.45±1.49
EEGNet	79.12±0.02	44.17±0.01	50.74±0.37	77.83±2.28	47.95±3.75	57.28±1.72	51.86±0.62	51.15±1.25	52.89±0.78
EEGInception	72.16±8.83	46.59±2.31	51.73±0.85	77.36±1.76	62.58±1.31	71.14±2.17	59.07±0.50	57.99±1.56	63.65±0.89
EEGConformer	74.74±0.65	54.40±0.46	61.01±1.45	81.31±0.67	69.64±1.35	79.58±1.21	<u>64.09±1.18</u>	<u>64.01±1.18</u>	69.83±1.58
BIOT	73.89±2.75	48.40±2.20	49.84±0.58	73.94±1.64	54.28±1.41	58.63±1.24	54.51±1.15	54.20±1.09	56.10±1.31
LaBraM	74.33±1.02	<u>55.90±1.13</u>	<u>61.95±1.26</u>	80.21±0.51	65.75±0.97	75.31±1.10	63.40±1.33	63.27±1.31	69.02±1.65
CBraMod	76.09±0.61	53.37±0.63	59.76±0.55	80.85±0.82	66.25±1.25	76.00±1.37	64.01±1.37	63.87±1.48	<u>69.97±1.40</u>

Datasets	mTBI-ODD (24,885 Trials) (96 Subjects, 3 Classes)			NSERP-MSIT (16,729 Trials) (42 Subjects, 4 Classes)			NSERP-ODD (27,865 Trials) (42 Subjects, 3 Classes)		
Metrics	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC
EEG Features	68.11±2.22	35.10±2.39	60.61±4.07	26.87±0.94	26.42±0.93	52.58±1.32	72.60±2.11	36.18±2.17	64.93±3.44
ERP Features	77.34±0.79	58.75±2.23	81.65±0.95	36.09±2.13	35.42±2.03	63.69±2.32	82.26±1.63	62.35±3.00	87.12±2.16
TCN	77.87±1.15	62.44±1.77	83.74±1.32	35.26±1.55	34.45±1.76	63.37±1.95	<u>84.17±1.25</u>	<u>66.54±2.54</u>	<u>90.17±1.68</u>
ModernTCN	79.16±0.98	63.77±1.33	<u>84.80±1.18</u>	36.94±2.38	36.72±2.37	65.04±2.63	83.28±0.97	63.21±2.68	88.82±1.32
TimesNet	78.09±1.28	63.66±2.68	84.30±1.35	36.86±2.40	36.44±2.24	65.31±2.47	82.85±0.94	63.54±2.03	88.74±1.30
PatchTST	78.25±1.64	63.57±1.91	84.49±1.25	36.59±2.58	36.22±2.67	64.13±2.87	82.94±1.01	63.70±1.89	88.34±1.66
iTransformer	78.42±1.13	<u>64.36±1.48</u>	84.54±1.40	36.40±2.40	36.03±2.34	64.68±2.51	81.74±1.36	61.85±2.43	87.52±1.69
Medformer	77.82±1.33	<u>63.99±1.37</u>	83.93±1.20	37.64±2.38	<u>37.29±2.45</u>	65.69±2.81	81.38±0.70	63.33±1.77	87.64±1.15
MedGNN	<u>78.51±0.97</u>	63.92±1.64	<u>84.74±1.05</u>	<u>37.74±2.26</u>	37.20±2.15	<u>66.40±2.55</u>	83.66±1.19	65.13±2.77	90.08±1.39
EEGNet	63.86±4.92	34.00±3.20	58.90±1.39	25.83±1.28	20.92±1.98	50.24±0.87	60.16±2.62	35.35±1.35	60.42±0.94
EEGInception	65.07±4.86	48.70±1.08	70.36±0.89	31.43±2.09	27.67±4.29	58.99±2.58	75.48±3.77	58.19±4.68	82.23±3.26
EEGConformer	<u>78.73±1.70</u>	65.79±2.15	85.63±1.45	<u>38.59±2.01</u>	<u>38.39±2.06</u>	<u>66.22±2.25</u>	<u>84.27±1.32</u>	68.40±2.25	90.53±1.43
BIOT	63.79±2.77	39.90±0.75	60.16±0.63	30.40±0.94	30.06±1.11	57.04±1.36	75.50±1.12	49.83±1.56	74.91±2.58
LaBraM	76.70±1.76	61.93±2.34	83.12±1.44	35.91±2.86	35.54±2.75	65.09±3.14	82.15±1.19	63.51±2.31	88.48±1.78
CBraMod	78.21±1.07	63.50±1.63	84.64±1.27	39.24±2.77	38.51±2.62	67.76±2.83	84.49±1.24	<u>67.42±2.98</u>	<u>90.40±1.61</u>

EEG segments and predicting their corresponding neural lexical entries) based on this lexicon. Equipped with learnable temporal and spatial embeddings, this model adaptively learns from EEG inputs of arbitrary channel counts and durations. We load their pre-trained weights for fine-tuning.

3) **CBraMod**: CBraMod [18] is a foundation model for EEG that adopts a criss-cross Transformer as its backbone. By employing parallel spatial and temporal attention mechanisms, it separately models dependencies between channels at the same time point and between time segments within the same channel, thereby better aligning with the inherent structure of EEG signals. Additionally, the model uses Asymmetric Conditional Position Embeddings (ACPE) to dynamically generate position embeddings, enabling it to adapt flexibly to EEG data with varying channel configurations and temporal durations. We load their pre-trained weights for fine-tuning.

D. Patch Embedding Comparison for ERP Transformer

Existing EEG Transformer models commonly employ three types of token embeddings: **multi-variate**, **uni-variate**, and

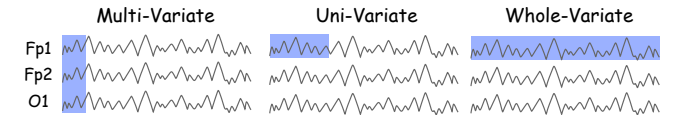


Fig. 3: **Transformer Embedding Comparison.** Three commonly used EEG Transformer patch embedding methods.

whole-variate. For example, Medformer [72] employs multi-variate patch embedding, CBraMod [18] adopts univariate patch embedding, and iTransformer [71] uses whole-variate patch embedding. Given a multivariate EEG input sample ($\mathbf{x} \in \mathbb{R}^{T \times C}$ and a patch length L , multi-variate embedding uses patches $\mathbf{p} \in \mathbb{R}^{L \times C}$, univariate embedding uses patches $\mathbf{p} \in \mathbb{R}^{L \times 1}$, and whole-variate embedding uses patches $\mathbf{p} \in \mathbb{R}^{1 \times C}$. These patches are then linearly projected into token embeddings that serve as inputs to the Transformer encoder. Figure 5 illustrates the three embedding strategies. To investigate which embedding method is most suitable for ERP data classification, we implement a test Transformer model with three different token embedding designs. All

TABLE III: **ERP-Based Disease Detection Results.** ERP-based brain disease detection focuses on classifying neurological disorders, such as Parkinson’s disease or ADHD. **Top-1**, **Top-2**, and **Top-3** results are highlighted in red, blue, and green.

Datasets	PD-SIM (55,921 Trials) (147 Subjects, 2 Classes)			PD-ODD (34,464 Trials) (145 Subjects, 2 Classes)			ADHD-WMRI (21,832 Trials) (59 Subjects, 2 Classes)		
	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC
EEG Features	64.32±2.30	57.02±3.23	64.41±4.44	67.68±3.36	61.23±3.32	70.01±3.42	58.43±3.07	55.76±3.61	56.65±6.30
ERP Features	63.76±3.54	58.78±2.57	66.41±3.71	69.29±2.60	63.65±2.71	72.05±2.50	59.79±4.79	57.40±5.93	59.48±10.23
TCN	64.63±3.53	58.76±5.73	64.92±5.32	66.07±2.35	59.43±2.81	68.50±2.28	62.74±2.46	60.57±2.82	67.05±3.76
ModernTCN	71.98±1.82	65.79±2.58	76.07±3.10	71.74±1.47	66.72±0.73	76.62±1.48	63.26±2.86	60.69±4.24	66.79±4.88
TimesNet	63.09±3.86	54.44±4.99	61.16±8.62	68.15±2.83	60.45±2.37	69.87±1.82	66.05±3.49	64.30±3.79	72.12±5.04
PatchTST	69.38±3.87	64.19±4.63	73.32±4.77	72.19±1.97	67.37±1.15	77.11±2.51	62.44±3.62	60.85±3.60	66.46±5.03
iTransformer	71.38±2.09	66.79±2.39	76.03±2.75	70.34±2.25	65.89±1.83	75.06±1.77	63.29±3.13	60.92±3.88	66.77±5.35
Medformer	68.62±1.59	61.28±3.68	70.95±2.71	68.22±2.29	61.61±2.49	72.10±2.81	64.48±2.60	61.47±3.25	68.69±3.91
MedGNN	57.17±4.72	49.88±3.69	56.94±6.91	64.61±3.83	58.15±4.21	68.59±3.61	62.70±3.98	59.97±5.71	66.44±7.18
EEGNet	66.08±3.17	64.17±2.84	71.63±3.45	65.25±4.48	61.84±3.59	70.46±5.57	52.70±8.00	44.77±5.24	43.99±5.10
EEGInception	58.90±1.55	52.26±5.35	57.52±8.47	63.73±1.85	60.01±2.36	65.97±1.88	65.22±3.62	63.12±4.59	69.60±5.73
EEGConformer	60.69±4.64	56.80±5.22	61.10±6.67	65.44±1.40	62.88±1.56	70.12±1.08	68.55±2.70	66.43±3.52	74.55±4.68
BIOT	64.17±4.40	56.04±5.45	62.37±6.41	65.74±3.58	59.76±5.30	69.03±3.47	54.29±5.62	53.05±5.73	54.61±8.30
LaBraM	63.95±5.29	57.29±4.90	63.47±6.11	67.82±5.22	62.55±3.78	71.62±7.46	63.82±9.56	62.11±9.61	67.99±13.05
CBraMod	63.03±2.87	57.01±3.05	64.73±3.72	66.01±0.95	60.41±4.12	71.06±2.71	61.98±5.24	59.96±5.40	65.24±7.45

Datasets	SCPD (10,224 Trials) (56 Subjects, 2 Classes)			RLPD (14,325 Trials) (56 Subjects, 2 Classes)			AOPD (9,830 Trials) (50 Subjects, 2 Classes)		
	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC	Accuracy	F1 Score	AUROC
EEG Features	61.32±2.28	61.21±2.28	66.63±3.19	60.87±3.44	60.43±3.40	62.70±4.09	61.83±3.32	61.43±3.55	67.33±4.17
ERP Features	57.55±5.36	57.41±5.27	61.93±8.01	54.55±3.88	53.42±3.92	58.89±6.35	61.43±2.73	60.82±3.08	67.16±3.30
TCN	70.26±4.83	70.15±4.75	77.31±6.78	68.67±2.67	68.40±2.59	73.23±6.78	66.11±6.75	65.76±6.81	70.57±8.43
ModernTCN	67.85±5.39	67.55±5.37	72.89±7.10	61.04±5.63	60.75±5.46	66.08±5.81	60.00±5.44	58.86±6.38	62.06±8.09
TimesNet	71.44±6.30	71.15±6.34	78.00±7.88	67.64±7.49	67.36±7.46	74.56±9.54	65.35±3.61	65.11±3.62	71.53±4.52
PatchTST	63.37±4.36	63.15±4.39	68.80±5.95	61.07±7.23	60.35±7.20	64.87±8.99	57.63±6.00	56.92±6.51	60.39±8.29
iTransformer	66.83±5.65	66.51±5.83	71.91±7.25	59.36±4.23	59.07±4.14	62.90±5.05	59.56±6.43	58.54±7.36	61.05±9.63
Medformer	65.22±6.80	64.93±6.96	71.91±7.87	63.45±6.43	63.18±6.20	69.80±8.35	64.94±6.59	64.59±6.60	70.56±8.78
MedGNN	66.13±9.71	65.73±9.97	72.34±12.06	64.68±6.65	64.36±6.80	72.76±8.73	62.93±3.93	62.48±4.00	69.29±5.56
EEGNet	64.78±4.61	63.57±4.96	68.41±5.85	63.18±1.77	62.53±1.61	67.12±2.79	62.51±7.17	60.41±8.64	64.92±9.70
EEGInception	74.48±5.53	74.33±5.42	82.79±6.36	70.76±4.43	70.57±4.42	77.32±6.43	69.51±6.59	69.26±6.52	77.32±8.81
EEGConformer	72.35±5.39	72.04±5.43	79.43±6.74	66.09±4.05	65.28±4.16	75.73±6.46	68.25±9.26	68.00±9.26	74.43±11.74
BIOT	70.00±4.05	69.42±4.61	76.71±6.79	65.72±9.04	65.09±8.85	69.29±12.06	62.15±8.73	61.57±8.80	66.51±10.06
LaBraM	72.27±4.17	71.71±4.30	79.46±6.52	74.47±6.34	73.87±6.94	84.82±5.72	62.05±9.25	61.55±9.20	67.40±13.80
CBraMod	67.33±3.45	66.63±3.89	73.34±8.45	63.28±9.41	62.65±9.14	66.26±12.52	65.99±3.78	65.28±3.88	71.43±4.80

other components of the Transformer architecture are kept identical across settings, allowing a controlled comparison of the embedding methods. We hope this comparison could inspire future works on designing Transformers for ERP analysis.

V. EXPERIMENTS

A. Experimental Setups

Batch sizes are 128 for all methods. The training epochs are set to 200 with early stopping (patience=15) based on the best F1 Score. We use AdamW with learning rates of 1×10^{-4} , scheduled by CosineAnnealingLR. Model performance is evaluated using Accuracy, F1 Score, and AUROC. We adopt Monte Carlo cross-validation under the **subject-independent** evaluation protocol [77], with a 6:2:2 train, validation, and test split. This setup ensures that no subject overlaps across splits while maintaining random subject partitioning across different seeds. For foundation model methods, including BIOT, LaBraM, and CBraMod, we fine-tune their publicly released checkpoints. Each experiment is repeated with five random seeds (41–45), and results are reported as the mean and standard deviation. All experiments are conducted on four NVIDIA RTX A5000

GPUs using Python 3.10 and PyTorch 2.5.1+cu121. Additional implementation details and training scripts for each method are provided in our GitHub repository.

B. ERP Stimulus Classification Results

We first investigate stimulus-type classification in ERP data, with results summarized in Table II. All experiments are conducted under a subject-independent evaluation setup, aiming to identify stimulus-related ERP patterns that generalize across subjects. The top-1, top-2, and top-3 performing methods are highlighted in red, blue, and green, respectively. Across the six datasets (CESCA-AOOD, CESCA-VOOD, CESCA-FLANKER, mTBI-ODD, NSERP-MSIT, and NSERP-ODD), the best method achieved F1 scores of 57.18%, 69.64%, 64.33%, 65.79%, 38.51%, and 68.40%, respectively. These results are substantially higher than random performance, indicating that meaningful and consistent stimulus-related ERP patterns can be extracted even under cross-subject evaluation. Among the three CESCA sub-datasets, the visual oddball and flanker paradigms exhibit higher discriminability than the auditory oddball paradigm. Since these datasets share the same group of subjects and recording conditions, this

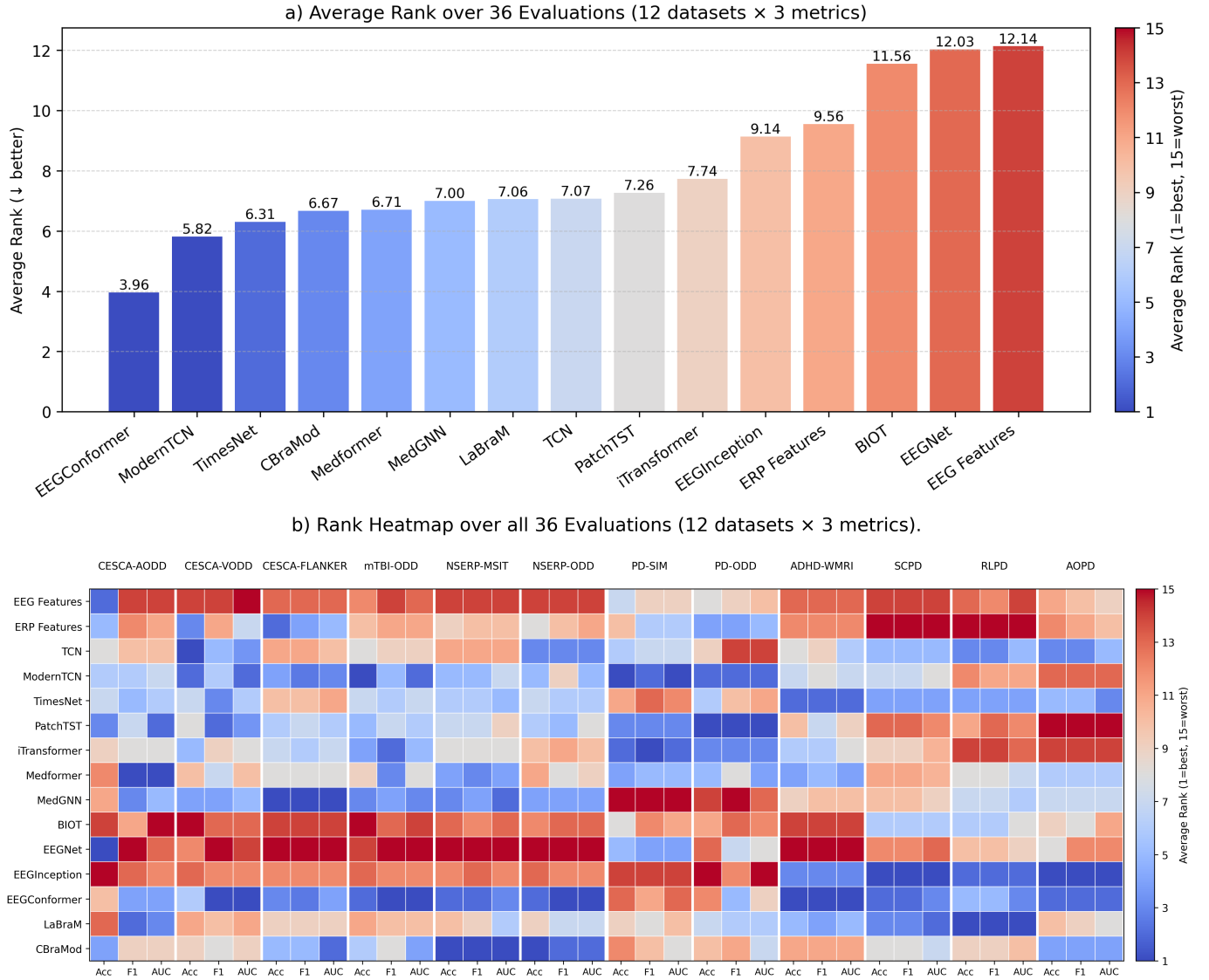


Fig. 4: **a)** Average performance rank of 15 methods across all 12 datasets and 3 evaluation metrics. For example, the value 3.96 for EEGConformer indicates an average rank of 3.96 over 36 evaluations. **b)** Rank heatmap of 15 methods across all 12 datasets and 3 evaluation metrics. **Lower ranks and deeper blue color indicate better performance** in both plots a and b.

observation suggests that visual oddball and flanker stimuli elicit more salient and distinguishable neural responses than passive auditory oddball stimuli. For the comparison of method performance, we observe that the top-tier results are predominantly achieved by deep learning models trained from scratch. In contrast, existing foundation models for medical time series and EEG do not demonstrate clear performance advantages when fine-tuned from their released checkpoints. Moreover, manually extracted features rarely achieve top-tier results. Notably, ERP-specific features consistently outperform generic EEG features in this task.

C. ERP-Based Disease Detection Results

We then investigate ERP-based disease detection performance, with results summarized in Table III. All experiments are conducted under a subject-independent evaluation setup, aiming to identify disease-specific ERP features that generalize

across subjects for real-world disease detection applications. Across the six datasets (PD-SIM, PD-ODD, ADHD-WMRI, SCPD, RLPD, and AOPD), the best method achieves F1 scores of 66.79%, 67.37%, 66.43%, 74.33%, 73.87%, and 69.26%, respectively. Overall, no single ERP paradigm consistently demonstrates a clear advantage for disease detection. Notably, the three smaller Parkinson’s disease datasets, SCPD, RLPD, and AOPD, achieve relatively higher performance than the two larger datasets, PD-SIM and PD-ODD. This trend may be attributed to the more balanced data distributions in the smaller datasets, which contain exactly equal numbers of Parkinson’s disease and healthy control subjects. For method comparison, top-tier performance is primarily achieved by deep learning models trained from scratch and by models fine-tuned from foundation model checkpoints. Two methods based on manually extracted features never achieve top-tier results. Besides, unlike ERP stimulus-type classification, ERP-specific

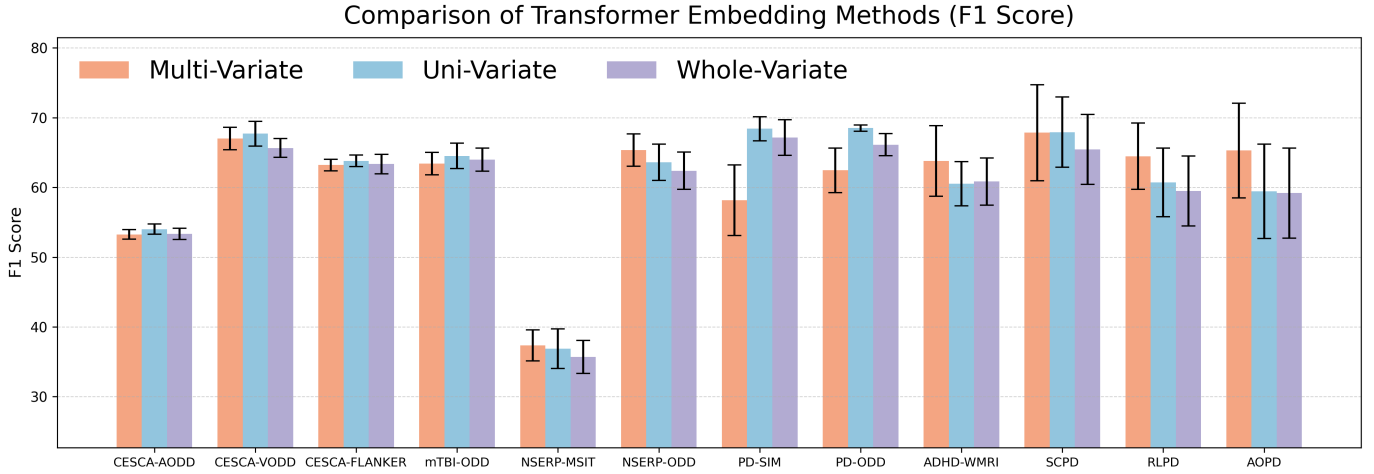


Fig. 5: F1 Score comparison among multi-variate, uni-variate, and whole-variate token embedding commonly used in existing EEG Transformer. All other Transformer components are kept identical.

features do not consistently outperform generic EEG features in disease detection. In several datasets, ERP-specific features perform worse, suggesting that the disease-related information in these datasets may be less tightly aligned with canonical ERP components.

D. Overall Performance Analysis

To provide a comprehensive comparison of different methods across multiple tasks and evaluation metrics, we further analyze the overall performance of 15 methods over **12 datasets** $\times$ **3 evaluation metrics**, resulting in **36 evaluations in total**. Figure 4 presents the average rank of each method across all 36 evaluations, along with a heatmap showing the individual ranks for each method on each evaluation. Lower average ranks and blue colors indicate better overall performance.

We observe that supervised deep learning models trained from scratch consistently outperform other methods in overall performance. Among all evaluated approaches, EEGConformer achieves the lowest average rank of 3.96, indicating the most stable and competitive performance across diverse tasks and evaluation metrics. Following EEGConformer, ModernTCN, TimesNet, CBraMod, and Medformer constitute the top-tier group. Notably, only CBraMod belongs to the foundation model category, while the remaining methods are trained from scratch. This suggests that existing foundation models can partially transfer to ERP classification tasks. One possible explanation is that current EEG foundation models are rarely pre-trained on ERP-specific data. Most pre-training resources consist of long-sequence spontaneous EEG and do not incorporate ERP-specific preprocessing steps, such as baseline correction or event-based trial epoching. As a result, the representations learned during pre-training may not be optimally aligned with ERP-centric downstream tasks. Manual feature-based methods consistently perform the worst overall. Between the two handcrafted feature methods, ERP-specific features perform relatively better than generic EEG features. These results highlight the limitations of handcrafted statistical and frequency-domain features, which struggle to maintain robust performance across diverse ERP

paradigms and disease conditions, particularly under subject-independent evaluation. This suggests that, beyond interpretability or neuroscience-oriented analysis, manual feature extraction offers limited advantages in real-world applications where end-to-end performance is the primary objective.

TABLE IV: **Patch Embedding Method Comparison.** Three commonly used token embedding methods in existing EEG Transformer models. All other Transformer components are kept identical. The number of parameters is comparable across the three methods. Results are reported in the F1 score.

Methods	Multi-Variate (0.864M)	Uni-Variate (0.818M)	Whole-Variate (0.822M)
Datasets			
CESC-A-ODD	53.26 $\pm$ 0.70	54.02$\pm$0.73	53.35 $\pm$ 0.81
CESC-A-VODD	67.00 $\pm$ 1.62	67.71$\pm$1.77	65.66 $\pm$ 1.33
CESC-A-FLANKER	63.21 $\pm$ 0.84	63.81$\pm$0.82	63.35 $\pm$ 1.39
mTBI-ODD	63.41 $\pm$ 1.62	64.51$\pm$1.83	63.98 $\pm$ 1.67
NSERP-MSIT	37.34$\pm$2.21	36.87 $\pm$ 2.85	35.69 $\pm$ 2.37
NSERP-ODD	65.36$\pm$2.32	63.59 $\pm$ 2.61	62.39 $\pm$ 2.69
PD-SIM	58.16 $\pm$ 5.05	68.41$\pm$1.73	67.14 $\pm$ 2.55
PD-ODD	62.45 $\pm$ 3.19	68.50$\pm$0.44	66.13 $\pm$ 1.58
ADHD-WMRI	63.77$\pm$5.06	60.53 $\pm$ 3.15	60.84 $\pm$ 3.37
SCPD	67.84 $\pm$ 6.87	67.91$\pm$5.04	65.44 $\pm$ 5.03
RLPD	64.47$\pm$4.74	60.73 $\pm$ 4.92	59.48 $\pm$ 5.00
AOPD	65.29$\pm$6.78	59.43 $\pm$ 6.76	59.19 $\pm$ 6.45

E. Patch Embedding Comparison for ERP Transformer

We compare three commonly used token embedding methods in existing EEG Transformer models to examine their suitability for ERP data classification. All other components, including the self-attention modules, feedforward networks, and layer normalization layers, are kept identical across methods. To ensure comparable model capacities, the patch length is set to 25 for the multi-variate and 100 for the uni-variate embedding methods. As a result, the total numbers of parameters are 0.864M, 0.818M, and 0.822M for the multivariate, univariate, and whole-variate token embedding methods, respectively.

All results are reported in terms of F1 score in Table IV. The uni-variate embedding method achieves the best performance on 7 out of 12 datasets, while the multi-variate embedding

method achieves the best results on the remaining 5 datasets. In contrast, the whole-variate embedding method does not achieve top performance on any dataset. These results demonstrate the advantage of uni-variate token embedding for ERP classification and suggest promising directions for designing ERP-specific Transformer backbones.

VI. DISCUSSION

In this paper, we conduct a comprehensive benchmark study on ERP data classification. Overall, performance on both stimulus-type classification and brain disease detection remains relatively limited for real-world deployment. This limitation may stem from a combination of dataset characteristics and methodological choices. Although the number of trials in existing ERP datasets is often sufficient for traditional manual feature-based analysis, it may still be inadequate for training data-hungry deep learning models. In addition, dataset quality factors, such as noise contamination and class imbalance, can further constrain achievable performance, which is consistent with our observations in Table III.

From a methodological perspective, EEGConformer, a model proposed several years ago, achieves the best average ranking in overall performance, outperforming more recent approaches, including foundation models. This observation suggests that recent advances in general EEG representation learning do not directly translate to the ERP sub-domain, highlighting the need for backbone architectures specifically tailored to ERP data characteristics. Our comparison in Table IV of token-embedding methods for the EEG Transformer may serve as a starting point for designing an ERP-specific Transformer method. Moreover, the lack of a clear advantage of existing foundation models over fully supervised training suggests that ERP-centric pre-training may also be necessary. Most current EEG foundation models are primarily pre-trained on spontaneous EEG corpora, such as TUEG [78], without incorporating ERP-specific preprocessing steps or pre-training strategies. In contrast, ERP data have time-locked trials and stimulus-specific temporal dynamics. These differences suggest that ERP-specific pre-training datasets and preprocessing pipelines are likely essential for improving downstream ERP classification performance.

VII. CONCLUSION

In this paper, we conduct a comprehensive benchmark study of two ERP tasks: ERP stimulus-type classification and ERP-based brain disease detection, across 12 datasets. We compare 15 methods, including two manual feature-extraction approaches, ten supervised deep learning methods, and three foundation models with pre-trained weights. In addition, to investigate the most suitable token embedding strategy for Transformer-based ERP classification, we conduct a controlled comparison of three commonly used patch embedding methods, aiming to inform the design of ERP-specific Transformer architectures. Based on the results from 36 evaluations, we summarize four key takeaways from this benchmark study. First, deep learning methods consistently outperform manual feature extraction approaches on ERP tasks. Second, existing EEG foundation models do not exhibit clear performance advantages

over supervised deep learning models trained from scratch. Third, EEGConformer achieves the most competitive overall performance among the evaluated methods for ERP classification. Fourth, uni-variate patch embedding demonstrates the strongest performance among the examined embedding strategies for ERP-specific Transformers. We hope that this benchmark study serves as a useful reference for method selection in ERP analysis and inspires future research on designing more effective ERP-oriented models for advanced ERP analysis and representation learning.

REFERENCES

- [1] Annmarie MacNamara, Keanan Joyner, and Julia Klawohn. Event-related potential studies of emotion regulation: A review of recent progress and future directions. *International Journal of Psychophysiology*, 176:73–88, 2022.
- [2] Estelle Pruvost-Robieux, Angela Marchi, Ilaria Martinelli, Eléonore Bouchereau, and Martine Gavaret. Evoked and event-related potentials as biomarkers of consciousness state and recovery. *Journal of Clinical Neurophysiology*, 39(1):22–31, 2022.
- [3] Truett Allison. Recording and interpreting event-related potentials. In *Cognitive psychophysiology: Event-related potentials and the study of cognition*, pages 1–36. Routledge, 2022.
- [4] Gregory A Light, Lisa E Williams, Falk Minow, Joyce Sprock, Anthony Rissling, Richard Sharp, Neal R Swerdlow, and David L Braff. Electroencephalography (eeg) and event-related potentials (erps) with human participants. *Current protocols in neuroscience*, 52(1):6–25, 2010.
- [5] Steven J Luck. Event-related potentials. 2012.
- [6] Randolph F Helfrich and Robert T Knight. Cognitive neurophysiology: Event-related potentials. *Handbook of clinical neurology*, 160:543–558, 2019.
- [7] Tracy DeBoer, Lisa S Scott, and Charles A Nelson. Methods for acquiring and analyzing infant event-related potentials. In *Infant EEG and event-related potentials*, pages 5–38. Psychology Press, 2013.
- [8] Xiang Zhang, Lina Yao, Xianzhi Wang, Jessica Monaghan, David McAlpine, and Yu Zhang. A survey on deep learning-based non-invasive brain signals: recent advances and new frontiers. *Journal of neural engineering*, 18(3):031002, 2021.
- [9] Shravani Sur and Vinod Kumar Sinha. Event-related potential: An overview. *Industrial psychiatry journal*, 18(1):70–73, 2009.
- [10] Zitao Fang, Chenxuan Li, Hongting Zhou, Shuyang Yu, Guodong Du, Ashwaq Qasem, Yang Lu, Jing Li, Junsong Zhang, and Sim Kuan Goh. Neuript: Foundation model for neural interfaces. *39th International Conference on Neural Information Processing Systems*, 2025.
- [11] Ruizhe Zheng, Lingyan Mao, Dingding Han, Tian Luo, Yi Wang, Jing Ding, and Yuguo Yu. Fapex: Fractional amplitude-phase expessor for robust cross-subject seizure prediction. *39th International Conference on Neural Information Processing Systems*, 2025.
- [12] Yihe Wang, Nadia Mammone, Darina Petrovsky, Alexandros T Tzallas, Francesco C Morabito, and Xiang Zhang. Adformer: A multi-granularity spatial-temporal transformer for eeg-based alzheimer detection. *arXiv preprint arXiv:2409.00032*, 2024.
- [13] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [14] Gemini Team, Petko Georgiev, Ving Ian Lei, Ryan Burnell, Libin Bai, Anmol Gulati, Garrett Tanzer, Damien Vincent, Zhufeng Pan, Shibo Wang, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530*, 2024.
- [15] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.
- [16] Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous eeg data in bci. *arXiv preprint arXiv:2405.18765*, 2024.
- [17] Guangyu Wang, Wenchao Liu, Yuhong He, Cong Xu, Lin Ma, and Haifeng Li. Eegpt: Pretrained transformer for universal and reliable representation of eeg signals. *Advances in Neural Information Processing Systems*, 37:39249–39280, 2024.

- [18] Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. Cbramod: A criss-cross brain foundation model for eeg decoding. *arXiv preprint arXiv:2412.07236*, 2024.
- [19] Ibtehaaj Hameed, Danish M Khan, Syed Muneeb Ahmed, Syed Sabeeh Aftab, and Hammad Fazal. Enhancing motor imagery eeg signal decoding through machine learning: A systematic review of recent progress. *Computers in Biology and Medicine*, 185:109534, 2025.
- [20] Katerina D Tzamourta, Nikolaos Giannakeas, Alexandros T Tzallas, Loukas G Astrakas, Theodora Afrantou, Panagiotis Ioannidis, Nikolaos Grigoriadis, Pantelis Angelidis, Dimitrios G Tsalikakis, and Markos G Tsipouras. Eeg window length evaluation for the detection of alzheimer's disease over different brain regions. *Brain sciences*, 9(4):81, 2019.
- [21] Katerina D Tzamourta, Theodora Afrantou, Panagiotis Ioannidis, Maria Karatzikou, Alexandros T Tzallas, Nikolaos Giannakeas, Loukas G Astrakas, Pantelis Angelidis, Evripidis Glavas, Nikolaos Grigoriadis, et al. Analysis of electroencephalographic signals complexity regarding alzheimer's disease. *Computers & Electrical Engineering*, 76:198–212, 2019.
- [22] NN Kulkarni and VK Bairagi. Extracting salient features for eeg-based diagnosis of alzheimer's disease using support vector machine classifier. *IETE Journal of Research*, 63(1):11–22, 2017.
- [23] Joseph N Mak, Dennis J McFarland, Theresa M Vaughan, Lynn M McCane, Phillippa Z Tsui, Debra J Zeitlin, Eric W Sellers, and Jonathan R Wolpaw. Eeg correlates of p300-based brain-computer interface (bci) performance in people with amyotrophic lateral sclerosis. *Journal of neural engineering*, 9(2):026014, 2012.
- [24] Xiaoou Li, Yuning Yan, and Wenshi Wei. Identifying patients with poststroke mild cognitive impairment by pattern recognition of working memory load-related erp. *Computational and Mathematical Methods in Medicine*, 2013(1):658501, 2013.
- [25] Bruce Wallace, Frank Knoefel, Rafik Goubran, Rocío A López Zunini, Zhaofen Ren, and Aaron Maccosham. Eeg/erp: Within episodic assessment framework for cognition. *IEEE Transactions on Instrumentation and Measurement*, 66(10):2525–2534, 2017.
- [26] Francisco J Fraga, Tiago H Falk, Paulo AM Kanda, and Renato Anghinah. Characterizing alzheimer's disease severity via resting-awake eeg amplitude modulation analysis. *PLoS one*, 8(8):e72240, 2013.
- [27] Luke Tait, George Stothart, Elizabeth Coulthard, Jon T Brown, Nina Kazanina, and Marc Goodfellow. Network substrates of cognitive impairment in alzheimer's disease. *Clinical Neurophysiology*, 130(9):1581–1595, 2019.
- [28] Markus Waser, Heinrich Garn, Reinhold Schmidt, Thomas Benke, Peter Dal-Bianco, Gerhard Ransmayr, Helena Schmidt, Stephan Seiler, Günter Sanin, Florian Mayer, et al. Quantifying synchrony patterns in the eeg of alzheimer's patients with linear and non-linear connectivity markers. *Journal of Neural Transmission*, 123:297–316, 2016.
- [29] Lucas R Trambaiolli, Ana C Lorena, Francisco J Fraga, Paulo AM Kanda, Renato Anghinah, and Ricardo Nitrini. Improving alzheimer's disease diagnosis with machine learning techniques. *Clinical EEG and neuroscience*, 42(3):160–165, 2011.
- [30] Magali T Schmidt, Paulo AM Kanda, Luis FH Basile, Helder Frederico da Silva Lopes, Regina Baratho, Jose LC Demario, Mario S Jorge, Antonio E Nardi, Sergio Machado, Jéssica N Ianof, et al. Index of alpha/theta ratio of the electroencephalogram: a new marker for alzheimer's disease. *Frontiers in aging neuroscience*, 5:60, 2013.
- [31] Xiaokun Liu, Chunlai Zhang, Zheng Ji, Yi Ma, Xiaoming Shang, Qi Zhang, Wencheng Zheng, Xia Li, Jun Gao, Ruofan Wang, et al. Multiple characteristics analysis of alzheimer's electroencephalogram by power spectral density and lempel-ziv complexity. *Cognitive neurodynamics*, 10:121–133, 2016.
- [32] Paulo Afonso Medeiros Kanda, Lucas R Trambaiolli, Ana C Lorena, Francisco J Fraga, Luis Fernando I Basile, Ricardo Nitrini, and Renato Anghinah. Clinician's road map to wavelet eeg as an alzheimer's disease biomarker. *Clinical EEG and neuroscience*, 45(2):104–112, 2014.
- [33] Heinrich Garn, Markus Waser, Manfred Deistler, Thomas Benke, Peter Dal-Bianco, Gerhard Ransmayr, Helena Schmidt, Guenter Sanin, Peter Santer, Georg Caravias, et al. Quantitative eeg markers relate to alzheimer's disease severity in the prospective dementia registry austria (prodem). *Clinical Neurophysiology*, 126(3):505–513, 2015.
- [34] Hamed Azami, Steven E Arnold, Saeid Sanei, Zhuoqing Chang, Guillermo Sapiro, Javier Escudero, and Anoopum S Gupta. Multiscale fluctuation-based dispersion entropy and its applications to neurological diseases. *IEEE Access*, 7:68718–68733, 2019.
- [35] Lucie Tylová, Jaromír Kukal, Václav Hubata-Vacek, and Oldřich Vyšata. Unbiased estimation of permutation entropy in eeg analysis for alzheimer's disease classification. *Biomedical Signal Processing and Control*, 39:424–430, 2018.
- [36] Shixin Peng, Ruyi Xu, Xin Yi, Xin Hu, Lili Liu, and Leyuan Liu. Early screening of children with autism spectrum disorder based on electroencephalogram signal feature selection with l1-norm regularization. *Frontiers in Human Neuroscience*, 15:656578, 2021.
- [37] Raja Majid Mehmood, Muhammad Bilal, S Vimal, and Seong-Whan Lee. Eeg-based affective state recognition from human brain signals by using hjorth-activity. *Measurement*, 202:111738, 2022.
- [38] Zhangjing Deng, Haoying Bai, Shijing Wu, Jiani Wu, Boyuan Xia, Yingxi Chen, Yurou He, Shuyu Li, and Yang Lü. Diagnosis of alzheimer's disease via machine learning approaches with integrated resting-state eeg and erp characteristics. *Cognitive Computation*, 17(6):166, 2025.
- [39] Vernon J Lawhern, Amelia J Solon, Nicholas R Waytowich, Stephen M Gordon, Chou P Hung, and Brent J Lance. Eegnet: a compact convolutional neural network for eeg-based brain-computer interfaces. *Journal of neural engineering*, 15(5):056013, 2018.
- [40] Eduardo Santamaria-Vazquez, Victor Martinez-Cagigal, Fernando Vaquerizo-Villar, and Roberto Hornero. Eeg-inception: a novel deep convolutional neural network for assistive erp-based brain-computer interfaces. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(12):2773–2782, 2020.
- [41] Qi Xin, Shaohai Hu, Shuaiqi Liu, Ling Zhao, and Yu-Dong Zhang. An attention-based wavelet convolution neural network for epilepsy eeg classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:957–966, 2022.
- [42] Jin Xie, Jie Zhang, Jiayao Sun, Zheng Ma, Liuni Qin, Guanglin Li, Huihui Zhou, and Yang Zhan. A transformer-based approach combining deep learning network and spatial-temporal information for raw eeg classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:2126–2136, 2022.
- [43] N Kasthuri, R Ramya, VS Arunprassath, S Abhineth, and S Bharathraj. Eeg conformer model based epileptic seizure prediction using deep learning. In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, pages 1–7. IEEE, 2024.
- [44] Di Wu, Siyuan Li, Jie Yang, and Mohamad Sawan. Neuro-bert: Rethinking masked autoencoding for self-supervised neurological pretraining. *arXiv preprint arXiv:2204.12440*, 2022.
- [45] Yi Ding, Neethu Robinson, Chengxuan Tong, Qiuhaio Zeng, and Cuntai Guan. Lgnet: Learning from local-global-graph representations for brain-computer interface. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7):9773–9786, 2023.
- [46] Jing Jin, Ruitian Xu, Ian Daly, Xueqing Zhao, Xingyu Wang, and Andrzej Cichocki. Moccn: A multiscale deep convolutional neural network for erp-based brain-computer interfaces. *IEEE Transactions on Cybernetics*, 54(9):5565–5576, 2024.
- [47] Yiyu Gui, MingZhi Chen, Yuqi Su, Guibo Luo, and Yuchao Yang. Eegmamba: Bidirectional state space model with mixture of experts for eeg multi-task classification. *arXiv preprint arXiv:2407.20254*, 2024.
- [48] Wei-Bang Jiang, Yansen Wang, Bao-Liang Lu, and Dongsheng Li. Neurolm: A universal multi-task foundation model for bridging the gap between language and eeg signals. *arXiv preprint arXiv:2409.00101*, 2024.
- [49] Berkay Döner, Thorir Mar Ingolfsson, Luca Benini, and Yawei Li. Luna: Efficient and topology-agnostic foundation model for eeg signal analysis. *39th International Conference on Neural Information Processing Systems*, 2025.
- [50] Yuchen Zhou, Jiamin Wu, Zichen Ren, Zhouheng Yao, Weiheng Lu, Kunyu Peng, Qihao Zheng, Chunfeng Song, Wanli Ouyang, and Chao Gou. Csbrian: A cross-scale spatiotemporal brain foundation model for eeg decoding. *39th International Conference on Neural Information Processing Systems*, 2025.
- [51] Carolin Breiting-Ziegler, Jana Tegelbeckers, Hans-Henning Flechtner, and Kerstin Krauel. Economical assessment of working memory and response inhibition in adhd using a combined n-back/nogo paradigm: An erp study. *Frontiers in human neuroscience*, 14:322, 2020.
- [52] JF Cavanagh. Eeg: 3-stim auditory oddball and rest in parkinson's. *OpenNeuro, OpenNeuro*, 2021.
- [53] Elif Isbell, Amanda N Peters, Dylan M Richardson, and Nancy E Rodas De León. Cognitive electrophysiology in socioeconomic context in adulthood. *Scientific Data*, 12(1):841, 2025.
- [54] James F Cavanagh, J Kevin Wilson, Rebecca E Rieger, Darbi Gill, James M Broadway, Jacqueline Hope Story Remer, Violet Fratzke, Andrew R Mayer, and Davin K Quinn. Erps predict symptomatic distress and recovery in sub-acute mild traumatic brain injury. *Neuropsychologia*, 132:107125, 2019.

- [55] Patrycja Dżianok, Ingrida Antonova, Jakub Wojciechowski, Joanna Dreszer, and Ewa Kublik. The nencki-symfonia electroencephalography/event-related potential dataset: Multiple cognitive tasks and resting-state data collected in a sample of healthy adults. *GigaScience*, 11:giac015, 2022.
- [56] Arun Singh, Rachel C Cole, Arturo I Espinoza, Jan R Wessel, James F Cavanagh, and Nandakumar S Narayanan. Evoked mid-frontal activity predicts cognitive dysfunction in parkinson’s disease. *Journal of Neurology, Neurosurgery & Psychiatry*, 94(11):945–953, 2023.
- [57] Darin R Brown, Sarah Pirio Richardson, and James F Cavanagh. An eeg marker of reward processing is diminished in parkinson’s disease. *Brain research*, 1727:146541, 2020.
- [58] Arun Singh, Sarah Pirio Richardson, Nandakumar Narayanan, and James F Cavanagh. Mid-frontal theta activity is diminished during cognitive control in parkinson’s disease. *Neuropsychologia*, 117:113–122, 2018.
- [59] Luca Pion-Tonachini, Ken Kreutz-Delgado, and Scott Makeig. Iclabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage*, 198:181–197, 2019.
- [60] Jing Wang, Yuxing Fang, Xiao Wang, Huichao Yang, Xin Yu, and Huali Wang. Enhanced gamma activity and cross-frequency interaction of resting-state electroencephalographic oscillations in patients with alzheimer’s disease. *Frontiers in aging neuroscience*, 9:243, 2017.
- [61] Raymundo Cassani, Tiago H Falk, Francisco J Fraga, Paulo AM Kanda, and Renato Anghinah. The effects of automated artifact removal algorithms on electroencephalography-based alzheimer’s disease diagnosis. *Frontiers in aging neuroscience*, 6:55, 2014.
- [62] Ruofan Wang, Jiang Wang, Shunan Li, Haitao Yu, Bin Deng, and Xile Wei. Multiple feature extraction and classification of electroencephalograph signal for alzheimers’ with spectrum and bispectrum. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 25(1), 2015.
- [63] Markus Waser, Manfred Deistler, Heinrich Garn, Thomas Benke, Peter Dal-Bianco, Gerhard Ransmayr, Dieter Grossegger, and Reinhold Schmidt. Eeg in the diagnostics of alzheimer’s disease. *Statistical Papers*, 54:1095–1107, 2013.
- [64] Lucie Tylova, Jaromir Kukal, and Oldrich Vysata. Predictive models in diagnosis of alzheimer’s disease from eeg. *Acta Polytechnica*, 53(2), 2013.
- [65] Aldo Mora-Sánchez, Gérard Dreyfus, and François-Benoît Vialatte. Scale-free behaviour and metastable brain-state switching driven by human cognition, an empirical approach. *Cognitive neurodynamics*, 13:437–452, 2019.
- [66] Robert J Barry, Sopa Kirkaikul, and Darren Hodder. Eeg alpha activity and the erp to target stimuli in an auditory oddball paradigm. *International journal of psychophysiology*, 39(1):39–50, 2000.
- [67] Colin Lea, Michael D Flynn, Rene Vidal, Austin Reiter, and Gregory D Hager. Temporal convolutional networks for action segmentation and detection. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 156–165, 2017.
- [68] Donghao Luo and Xue Wang. Modernrtn: A modern pure convolution structure for general time series analysis. In *The twelfth international conference on learning representations*, pages 1–43, 2024.
- [69] Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. Timesnet: Temporal 2d-variation modeling for general time series analysis. *arXiv preprint arXiv:2210.02186*, 2022.
- [70] Y Nie. A time series is worth 64words: Long-term forecasting with transformers. *International conference on learning representations*, 2023.
- [71] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting. *International conference on learning representations*, 2023.
- [72] Yihe Wang, Nan Huang, Taida Li, Yujun Yan, and Xiang Zhang. Medformer: A multi-granularity patching transformer for medical time-series classification. *Advances in Neural Information Processing Systems*, 37:36314–36341, 2024.
- [73] Wei Fan, Jingru Fei, Dingyu Guo, Kun Yi, Xiaozhuang Song, Haolong Xiang, Hangting Ye, and Min Li. Towards multi-resolution spatiotemporal graph learning for medical time series classification. In *Proceedings of the ACM on Web Conference 2025*, pages 5054–5064, 2025.
- [74] François Chollet. Xception: Deep learning with depthwise separable convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1251–1258, 2017.
- [75] Yonghao Song, Qingqing Zheng, Bingchuan Liu, and Xiaorong Gao. Eeg conformer: Convolutional transformer for eeg decoding and visualization. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31:710–719, 2022.
- [76] Chaoqi Yang, M Westover, and Jimeng Sun. Biot: Biosignal transformer for cross-data learning in the wild. *Advances in Neural Information Processing Systems*, 36:78240–78260, 2023.
- [77] Yihe Wang, Nan Huang, Nadia Mammone, Marco Cecchi, and Xiang Zhang. Lead: Large foundation model for eeg-based alzheimer’s disease detection. *arXiv preprint arXiv:2502.01678*, 2025.
- [78] Iyad Obeid and Joseph Picone. The temple university hospital eeg data corpus. *Frontiers in neuroscience*, 10:196, 2016.